

Computer Architecture

Problem Set #1

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This problem set is **ungraded and not collected**. We will release solutions two weeks after the problem set is released. Please stop by office hours if you have questions.

1. Compute the Clocks Per Instruction (CPI) of a machine which has an average CPI for ALU operations of 1.1, a CPI for branches/jumps of 3.0, and a hit rate of 60% in the cache. A hit in the cache takes 1 cycle pipelined and a cache miss takes 120 cycles. Assume 22% of instructions are loads, 12% are stores, 20% are branches/jumps and the balance are ALU operations.
2. You are a processor designer and have to make a decision between building a processor which executes at 1GHz and has an average CPI 1.2 and a processor which executes at 2GHz, but has a CPI of 2. Which is better to build and why?
3. For the following RISC-V code snippet, identify all of the RAW, WAW, and WAR hazards. Provide a list for each hazard. Hint: remember that you have to check more than just neighboring instructions.

```
add x1, x2, x3
sub x3, x4, x6
mul x5, x4, x7
addi x5, x5, 1
sub x6, x3, x9
andi x2, x1, x9
```

4. For each of the characteristics below, complete the table by checking the appropriate entry (Architecture or Implementation) to indicate whether each feature is most often a characteristic of a computer architecture or of a particular CPU implementation.

Characteristic	Architecture?	Implementation?
# of bits in an instruction		
# of registers visible to programmers		
# of floating-point divider units		
RISC vs. CISC		
# of instructions issued per cycle		
Clock rate of the processor		
Pipeline depth		

5. Page B-60 in H&P6, Problem B.2 a,b
6. How many SRAM bits are required to implement an 8KB two-way set-associative cache with a 64-byte block size? Assume that each cache line includes a single valid bit and excludes dirty bits. Additionally, consider there is one bit per set for true LRU implementation. Assume that the address size of the machine is 32-bits and that the machine allows for byte addressing.
7. Using figure B.9 on page H&P6 B-25 and table B.8 on page B-24. Which has a lower miss rate, a 256KB direct mapped cache or a 64-KB 8-way cache? Which of the three C's drives the previous result?